CSCE 411: Design and Analysis of Algorithms (Honors)

Week Lecture 2: Markov Chains and PageRank

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1 Web Search and PageRank

Suppose a search engine finds one million pages containing the word `university`. Which pages should it display first?
Text matching tells us whether a page mentions a topic. The web graph provides another kind of information: which pages _____

Counting incoming links One idea is to rank a page by the number of other pages that link to it. What could go wrong with this rule?

A recursive notion of importance Instead, suppose every page divides its importance among its outgoing links.

- Links from important pages should contribute _____
- A page with many outgoing links should contribute _____

This definition is circular. To determine which pages are important, we seem to need to know which pages are important already.

Question 1. *How might we make such a circular definition precise?*

A *Circular definitions can never be useful*

B *Write equations that all the scores must satisfy together*

C *Choose the most important page in advance*

We will use a random walk to define the scores, then analyze and compute them using Markov chains.

2 Markov Chains

A **Markov chain** is a random sequence of states $X_0, X_1, X_2, \ldots$. Given the present state, the next-state distribution does not depend on the earlier history:

$$\Pr[X_{t+1} = i \mid X_t = j, X_{t-1}, \ldots, X_0] = \Pr[X_{t+1} = i \mid X_t = j].$$

We consider histories with positive probability, and assume that the transition rule is the same at every step.

The transition matrix Throughout this lecture, distributions are column vectors, and

$$P_{ij} = \Pr[X_{t+1} = i \mid X_t = j].$$

Column j describes where the chain goes *from* state j. Thus,

$$P_{ij} \geq 0, \qquad \sum_{i=1}^{n} P_{ij} = __.$$

We call this a **column-stochastic** matrix.

Following the probability mass Let $p_i^{(t)} = \Pr[X_t = i]$. How much mass moves from state j to state i in the next step?

_____.

Adding the contributions from all possible starting states gives

$$p_i^{(t+1)} = _____, \qquad p^{(t+1)} = _____.$$

Question 2. *If the chain starts at state j with probability 1, where do we find its distribution after one step?*

A *Row j of P*

B *Column j of P*

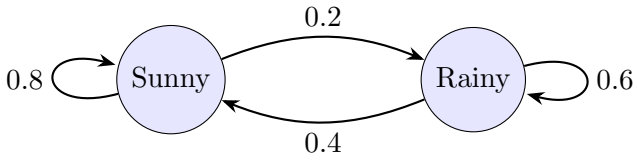
C *The diagonal of P*

A two-state weather model

Suppose tomorrow's weather depends only on today's weather:

- After a sunny day, tomorrow is sunny with probability 0.8.
- After a rainy day, tomorrow is sunny with probability 0.4.

This is a simplified model. Real weather may require more information than these two states.



Write the matrix In the order (Sunny, Rainy), fill in the transition matrix:

$$P = \begin{pmatrix} ____ & ____ \\ ____ & ____ \end{pmatrix}.$$

Check that each _____ sums to 1.

Predict two days ahead If today is sunny, then $p^{(0)} = (1, 0)^T$ and $p^{(1)} = (0.8, 0.2)^T$. What is the probability that it is sunny two days from now?

Why are Markov chains useful? They turn a local transition rule into predictions about later behavior. Similar models describe queues, movement through a network, or changes between operating states of a machine.

Multiplying by P tracks the distribution. Simulating the chain produces one particular sequence of states. These are different objects.

3 Stationary Distributions

A probability vector π is a **stationary distribution** if

$$P\pi = \pi, \qquad \pi_i \geq 0, \qquad \sum_i \pi_i = 1.$$

If we start with distribution π, the distribution after one step is _____

Return to the weather model Write $\pi = (s, 1-s)^T$. The sunny coordinate must satisfy

$$s = 0.8s + 0.4(1-s).$$

Solve for s:

Question 3. *What does stationarity mean for this weather model?*

A *Sunny days can no longer be followed by rainy days*

B *The probability of each weather state stays the same*

C *Successive days are independent*

Three different questions

1. Does a stationary distribution *exist*?
2. Is it *unique*?
3. Does $P^t p^{(0)}$ *converge* to it from every start?

Every finite stochastic matrix has a stationary distribution. We will use this fact without proof. The other two properties need additional arguments.

What about one long sample path? For a finite irreducible chain, the fraction of time spent in state i converges to π_i with probability 1. Here irreducible means that every state can reach every other state. This long-run frequency statement does not require consecutive observations to be independent.

Random walks and sampling

Consider a connected undirected graph with at least one edge. From vertex j, choose a neighbor uniformly. Then

$$P_{ij} = \begin{cases} 1/\deg(j), & \text{if } \{i,j\} \in E, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 3.1. *The vector $\pi_i = \deg(i)/(2|E|)$ is a stationary distribution.*

First, check the normalization Why do the entries of π sum to 1?

Next, check one coordinate For a vertex i,

$$(P\pi)_i = \sum_{j:\{i,j\}\in E} \frac{1}{\deg(j)} \frac{\deg(j)}{2|E|} = \underline{\qquad\qquad} = \pi_i.$$

Each neighbor contributes the same amount, and there are _____

Example: A star A center is connected to four leaves. What stationary probability does each vertex receive?

$$\pi_{\text{center}} = _____, \qquad \pi_{\text{each leaf}} = _____.$$

Question 4. *A researcher samples users by repeatedly moving to a uniformly chosen friend. Is this a uniform sample of users?*

A *Yes, because every choice is uniform*

B *No, users with more friends have larger stationary probabilities*

C *No, users with fewer friends have larger stationary probabilities*

Random walks let us sample or explore a large graph using only local information. We must check which distribution the walk actually samples.

4 BrokenRank (Almost PageRank)

Let $G = (V, E)$ be the directed graph of webpages, with $|V| = n$. Define $L_{ij} = 1$ if $j \to i$, and 0 otherwise. Let

$$m_j = \sum_i L_{ij}$$

be the number of outgoing links from page j.
For now, assume every page has at least one outgoing link. A random surfer follows an outgoing link uniformly, so

$$P_{ij} = \underline{\qquad\qquad}.$$

How should a page receive importance? If page j has importance p_j, how much does it send along each outgoing link?

Our preliminary ranking asks for a probability vector satisfying

$$p_i = \underline{\qquad\qquad\qquad}, \qquad p = \underline{\qquad}.$$

This is precisely a stationary distribution of the surfer's walk.

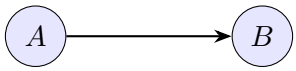
GenericRandomWalk$(G, p^{(0)})$
  Draw the starting page j from $p^{(0)}$
  while the surfer continues **do**
    Choose an outgoing edge (j, i) uniformly
    Set $j \leftarrow i$
  end while

A small calculation A page has current mass 0.12 and three outgoing links. It sends _____ along each link, so its total outgoing contribution is _____

The equations look reasonable. What happens when the graph has dead ends, closed groups of pages, disconnected components, or cycles?

Dead ends and spider traps

A page with no outgoing links Suppose the only link is $A \to B$, and B has no outgoing links.



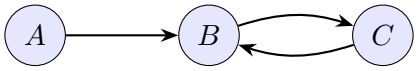
We call B a **dangling node**. What goes wrong with the formula $P_{ij} = L_{ij}/m_j$?

Suppose we simply replace the column for B by zeros. Starting at A, the distributions would become

$$\begin{pmatrix}1\\0\end{pmatrix}, \quad \begin{pmatrix}0\\1\end{pmatrix}, \quad _____ .$$

The last vector is not a probability distribution: its entries sum to _____

A group with no links to the outside Now suppose the links are $A \to B$, $B \to C$, and $C \to B$.



Once the surfer enters $\{B, C\}$, it _____ Such a closed group is called a **spider trap**.
Why might this give an undesirable ranking?

Existence, uniqueness, and convergence can differ

Disconnected components Suppose pages $1, 2, 3$ form a directed cycle, and pages $4, 5$ form a separate directed cycle. Both vectors below are stationary:

$$\pi^{(1)} = (1/3, 1/3, 1/3, 0, 0)^T, \qquad \pi^{(2)} = (0, 0, 0, 1/2, 1/2)^T.$$

Why is $a\pi^{(1)} + (1-a)\pi^{(2)}$ also stationary for every $0 \le a \le 1$?

There is no unique stationary ranking. The amount of mass in each component depends on _____

A strongly connected graph can still oscillate For the two-page cycle $1 \to 2 \to 1$,

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \pi = \begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}.$$

The stationary distribution is unique. But from $(1, 0)^T$, the walk alternates between $(1, 0)^T$ and $(0, 1)^T$, so its distribution _____

Question 5. *Does the fraction of visits to page* 1 *still approach* $1/2$*?*

A *Yes, because the sample path alternates*

B *No, because the time-t distribution oscillates*

A standard finite-chain theorem A chain is **irreducible** if every state can reach every other state. An irreducible chain is **aperiodic** if return times have greatest common divisor 1. If a finite chain has both properties, its stationary distribution is unique and the time-t distribution converges to it from every start.

5 PageRank

We will modify the surfer so that it can leave a page even when there are no links, and can escape a closed group of pages.

First, repair dangling nodes Choose a probability vector v, called the **restart distribution**. For ordinary PageRank, $v_i = 1/n$. Define

$$P_{ij} = \begin{cases} L_{ij}/m_j, & m_j > 0, \\ v_i, & m_j = 0. \end{cases}$$

At a dangling node, the surfer draws a new page from v. Each column now sums to ____

Next, allow teleportation Fix $0 < d < 1$. At each step,

- With probability d, take a step according to P.
- With probability $1 - d$, draw a fresh page from v.

Thus d is the link-follow probability; the teleportation probability is _____ A common illustrative choice is $d = 0.85$.
Let $\mathbf{1}$ be the all-ones column vector. The new transition matrix is

$$Q = dP + (1 - d)v\mathbf{1}^T.$$

Since $\mathbf{1}^T p = 1$ for a probability vector p,

$$Qp = _____.$$

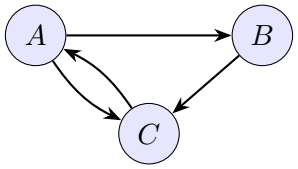
A PageRank vector therefore satisfies

$$\pi = Q\pi = _____.$$

Why does this help? For uniform v, every entry of Q is at least $(1 - d)/n > 0$. Every page can reach every other page, and every page has a self-loop. Thus the chain is _____

Compute PageRank on three pages

Consider the links $A \to B$, $A \to C$, $B \to C$, and $C \to A$.



Use $d = 1/2$ and $v = (1/3, 1/3, 1/3)^T$. In the order (A, B, C),

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1/2 & 0 & 0 \\ 1/2 & 1 & 0 \end{pmatrix}, \qquad p^{(t+1)} = \tfrac{1}{2} P p^{(t)} + \tfrac{1}{2} v.$$

Take two steps Starting from $p^{(0)} = v$, compute $p^{(1)}$ and $p^{(2)}$.

Solve for the stationary scores The fixed-point equations are

$$\pi_A = \tfrac{1}{6} + \tfrac{1}{2}\pi_C, \qquad \pi_B = \tfrac{1}{6} + \tfrac{1}{4}\pi_A, \qquad \pi_C = \tfrac{1}{6} + \tfrac{1}{4}\pi_A + \tfrac{1}{2}\pi_B.$$

Together with normalization, they give

$$\pi = \underline{\qquad\qquad\qquad\qquad\qquad\qquad}.$$

Check the equations and compare the ranking with the two-step calculation.

6 Computing PageRank

Although Q is dense, we do not need to construct it. Teleportation contributes only a multiple of v.
Let $z = \sum_{j:m_j=0} p_j$ be the current mass at dangling nodes. Then

$$p_i^{\text{new}} = ((1-d) + dz)v_i + d \sum_{j:j\to i} \frac{p_j}{m_j}.$$

Where do the terms $1-d$ and dz come from?

PageRank(G, d, v, ϵ), where $0 < d, \epsilon < 1$
  Compute the out-degrees m_j and initialize $p \leftarrow v$
  Set $T \leftarrow \lceil \log(2/\epsilon)/(1-d) \rceil$
  for $t = 1, \ldots, T$ **do**
    $z \leftarrow \sum_{j:m_j=0} p_j$
    $p_i^{\text{new}} \leftarrow ((1-d) + dz)v_i$ for all i
    for each edge $(j, i) \in E$ **do**
      $p_i^{\text{new}} \leftarrow p_i^{\text{new}} + d\, p_j/m_j$
    end for
    $\Delta \leftarrow \sum_i |p_i^{\text{new}} - p_i|$
    $p \leftarrow p^{\text{new}}$
    if $\Delta \le (1-d)\epsilon$ **then**
      return p
    end if
  end for
  return p

Why use a separate array? What could go wrong if we modified p while scanning the edges?

We will justify both the fixed budget T and the early stopping rule. The analysis assumes exact arithmetic.

Running time and dangling nodes

Question 6. *Using adjacency lists, what is the running time of one iteration?*

A $O(\log n)$

B $O(n + |E|)$

C $O(n^2)$, *because* Q *is dense*

D $O(n|E|)$

With t iterations, the running time is _____ Storage is $O(n+|E|)$ including the graph, with $O(n)$ extra space for the vectors.

Return to the two-page dead end The only edge is $A \to B$, and B is dangling. Let $d = 1/2$, $v = (1/2, 1/2)^T$, and $p = (1/2, 1/2)^T$.

- The dangling mass is $z =$ _____.
- The shared restart contribution to each page is

$$((1 - d) + dz)/2 = _____.$$

- The edge $A \to B$ contributes an additional _____ to B.

Therefore,

$$p^{\text{new}} = _____.$$

What vector would we get if we omitted dangling redistribution? Would it sum to 1?

In general, the restart part contributes $(1 - d) + dz$ and the ordinary edges contribute $d(1 - z)$. Their sum is ____

7 Why Does the Iteration Converge?

For a vector x, write $\|x\|_1 = \sum_i |x_i|$. The distance between two probability vectors is at most ____ Half of this distance is called total variation distance.

First, understand the intuition Imagine two surfers with different starts. Use the same teleportation decisions for both. When they teleport, send them to the same draw from v and use identical random choices afterward.
What is the probability that no teleportation occurs in the first t steps?

___.

Teleportation makes the walk forget its starting point. We will now express this as an inequality.

Lemma 7.1. *If P is column-stochastic, then $\|Px\|_1 \leq \|x\|_1$ for every vector x.*

Use the triangle inequality

$$\|Px\|_1 = \sum_i \left| \sum_j P_{ij} x_j \right| \leq \sum_i \sum_j P_{ij} |x_j| = \sum_j |x_j| \sum_i P_{ij}.$$

Each column sum is ___, so the last expression is _____

Now compare two PageRank updates For probability vectors p, q, their restart terms cancel:

$$Qp - Qq = _____.$$

Applying the lemma gives

$$\|Qp - Qq\|_1 \leq _____.$$

Thus each step shrinks discrepancies by a factor at most ____

Turn the inequality into a guarantee

Theorem 7.2. *For $0 < d < 1$ and any restart distribution v, PageRank has a unique stationary probability vector π. From every initial probability vector,*

$$\|p^{(t)} - \pi\|_1 \le d^t \|p^{(0)} - \pi\|_1 \le 2d^t.$$

Existence and uniqueness Existence follows because Q is a finite stochastic matrix. If π and ρ were both stationary, contraction would give

$$\|\pi - \rho\|_1 = \|Q\pi - Q\rho\|_1 \le d\|\pi - \rho\|_1.$$

Since $d < 1$, this forces _____ Applying contraction repeatedly with $q = \pi$ proves the error bound.

How many iterations suffice? To make $2d^t \le \epsilon$, it is enough that

$$t \ge \frac{\log(2/\epsilon)}{-\log d}.$$

Since $-\log d \ge 1 - d$, our budget $T = \lceil \log(2/\epsilon)/(1-d) \rceil$ suffices. The total running time is

_____ .

When can we stop early? Let $p^{\text{new}} = Qp$ and $\Delta = \|p^{\text{new}} - p\|_1$. By the triangle inequality and contraction,

$$\|p - \pi\|_1 \le \Delta + d\|p - \pi\|_1, \qquad \|p - \pi\|_1 \le _____ .$$

The new vector has error at most $d\Delta/(1-d)$. Thus stopping when $\Delta \le (1-d)\epsilon$ is also valid. The budget T ensures the stated runtime even if early stopping never occurs.

8 Personalized PageRank

Why must the restart distribution be uniform? Suppose we instead put more restart probability on pages related to a topic or a user's interests.
The equation stays the same:

$$\pi = dP\pi + (1 - d)v.$$

The scores now measure importance relative to the chosen restart distribution. In an item graph, restarting at a user's seed items can help identify related items.

What if some entries of v are zero? The chain may no longer be irreducible. Does the contraction calculation still work?

Choosing the link-follow probability

- A smaller d gives more weight to _____
- A larger d permits longer stretches of link-following, but the convergence guarantee becomes _____
- At $d = 0$, the answer is $\pi =$ __. At $d = 1$, teleportation disappears and Broken-Rank's issues can return.

Why does π approach v as $d \to 0$? For a fixed repaired matrix P, use the fixed-point equation to show that $\|\pi - v\|_1 \leq 2d$.

PageRank describes a chosen random-surfer model. A large score does not certify truth or relevance to every query, and links can be manipulated. A search system can use PageRank alongside textual relevance and other signals.

9 What Have We Learned?

Question 7. *A chain has a unique stationary distribution. What must follow?*

A *The chain eventually stops moving*

B *The time-t distribution converges from every start*

C *Neither statement must follow*

Question 8. *Why can we compute a PageRank update in $O(n + |E|)$ time even though Q is dense?*

A *Most teleportation entries are actually zero*

B *The teleportation contribution is a multiple of v*

C *We ignore teleportation during the computation*

Complete the three equations

$p^{(t+1)} = _____$, $\pi = _____$, $\|p^{(t)} - \pi\|_1 \leq _____$.

Check the modeling choices Why does adding teleportation to an unrepaired zero dangling column still fail to produce a stochastic matrix?

Where else could we use a random walk? Suggest an application in which a local transition rule could define a useful global score or sampling distribution. What would you need to check?

10 Optional: PageRank as a Random-Length Walk

Start with the fixed-point equation and substitute it into itself:

$$\pi = (1-d)v + dP\pi = (1-d)v + d(1-d)Pv + d^2P^2\pi.$$

If we continue, the remaining term tends to zero in ℓ_1, giving

$$\pi = (1-d)\sum_{k=0}^{\infty} d^k P^k v.$$

Why can we view this as a mixture of probability distributions?

SamplePageRank(P, d, v)
  Draw a starting page j from v
  while true **do**
    if an independent coin is heads, with probability $1-d$ **then**
      return j
    end if
    Take one step from j according to the repaired matrix P
  end while

How many steps do we take? Let K be the number of steps before stopping. Then

$$\Pr[K = k] = _____, \qquad k = 0, 1, 2, \ldots.$$

Conditioned on $K = k$, the endpoint distribution is $P^k v$. Therefore the unconditional endpoint distribution is ____
What is $\mathbb{E}[K]$? What does it tell us about the effect of d?

Power iteration propagates all the probability mass. Independent random-length walks sample from the same target distribution.

11 Optional: Laziness and Sampling Accuracy

Can we remove periodicity without changing stationarity? For a connected undirected graph with at least one edge, consider

$$P_{\text{lazy}} = \tfrac{1}{2}I + \tfrac{1}{2}P.$$

The walker stays put with probability 1/2 and otherwise takes an ordinary step. If $P\pi = \pi$, then

$$P_{\text{lazy}}\pi = \tfrac{1}{2}\pi + \tfrac{1}{2}P\pi = __.$$

The self-loops remove periodicity, while the degree-proportional stationary distribution stays the same.
Why would laziness fail to connect disconnected components?

Estimate one score using independent samples Run SamplePageRank independently R times. Let $Y_r = 1$ if run r ends at page i, and 0 otherwise. Define

$$\widehat{\pi}_i = \frac{1}{R}\sum_{r=1}^{R} Y_r, \qquad \mathbb{E}[\widehat{\pi}_i] = ___.$$

Hoeffding's inequality gives

$$\Pr[|\widehat{\pi}_i - \pi_i| > \eta] \le 2e^{-2R\eta^2}.$$

How large should R be to make this probability at most δ?

$$R \ge _____.$$

For all n pages simultaneously, a union bound replaces $\log(2/\delta)$ by _____

Why insist on independent runs? Could we apply the same argument directly to consecutive positions on one long walk?